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Tutorial 5

DSAI6707: Advanced Computational Intelligence

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Outline

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1. Multi-objective optimization and Pareto dominance
2. Multi-objective evolutionary algorithms
3. MOGA: algorithm, exercise and solution
4. NSGA-II: algorithm, exercise and solution
5. Comparison and conclusions

Multi-objective problems

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A multi-objective optimization problem (MOP) has at least two objectives.

$$\text{minimize} \quad f(x) = [f_1(x), f_2(x), \ldots, f_m(x)]$$

$$\text{subject to} \quad x \in \Omega, \qquad m \geq 2$$

x: decision variables Ω: feasible set

The objectives may conflict, so several trade-off solutions can be useful.

Conflicting objectives

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Example: choosing a model for deployment

Model	Latency (ms) ↓	Error (%) ↓
A	10	12
B	20	8
C	35	5
D	25	10

A is faster than B, but B is more accurate.

B improves both objectives relative to D.

Pareto dominance

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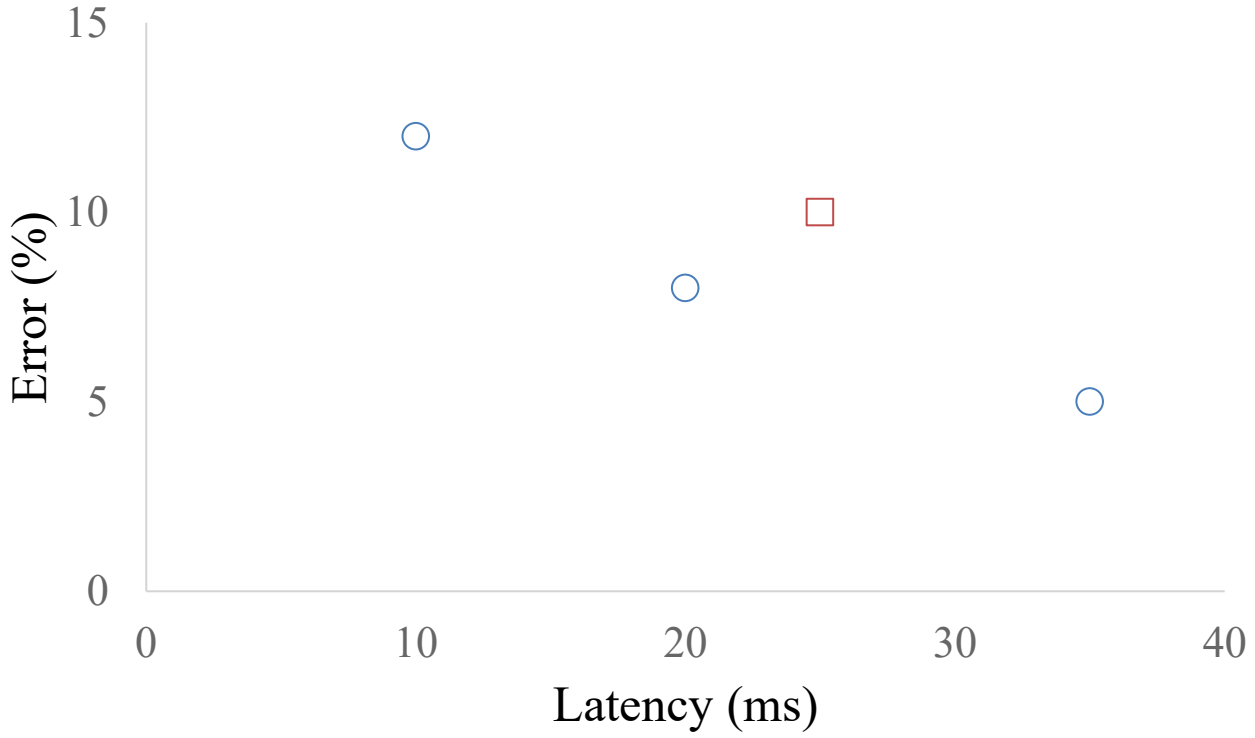
For minimization, a dominates b ($a \prec b$) when:

$f_j(a) \leq f_j(b)$ for every objective j

and $f_j(a) < f_j(b)$ for at least one objective.

Comparison	Conclusion
B = (20, 8), D = (25, 10)	B dominates D
A = (10, 12), B = (20, 8)	Neither dominates the other

Pareto set and front



Red: non-dominated Blue: dominated

In this four-model example, A, B and C are non-dominated.

Pareto set

Feasible decisions that no other feasible decision dominates.

Pareto front

The objective vectors of the Pareto-optimal decisions.

Evolutionary search

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Multi-objective Evolutionary Algorithms (MOEAs)

A population explores several trade-offs in one search run.

Selection encourages convergence towards better objective values.

Diversity preservation maintains alternatives across the front.

MOGA and NSGA-II are genetic-algorithm-based MOEAs.

The MOEA cycle

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1. Initialize a population of candidate decisions.
2. Evaluate every objective for each candidate.
3. Select parents using objective quality and diversity.
4. Apply crossover and mutation to generate offspring.
5. Form the next population and repeat until termination.

Output: a non-dominated approximation set for subsequent decision making.

MOGA: algorithm

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Multi-objective Genetic Algorithm (Fonseca–Fleming, 1993)

1. Initialize and evaluate the population.
2. Count dominators and assign a rank to each individual.
3. Assign rank-based fitness and apply fitness sharing.
4. Select parents in proportion to their adjusted fitness.
5. Generate and evaluate offspring using crossover and mutation.
6. Replace the population and repeat until the stopping criterion.

MOGA: rank and fitness

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Dominance-count rank

$$r(x) = 1 + |\{y \in P : y \prec x\}|$$

A non-dominated individual has rank 1. Lower ranks are better.

Fitness assignment

1. Sort individuals by rank and assign decreasing position scores.

2. Average scores within each rank to obtain equal base fitness.

MOGA: fitness sharing

Sharing reduces the reproductive advantage of crowded solutions.

$$n_i = \Sigma_j \, sh(d_{ij}) \qquad \text{and} \qquad F'_i = F_i / n_i$$

$$sh(d) = \max(0, 1 - d / \sigma)$$

n_i measures crowding among individuals of the same rank, including itself.

Rescale within each rank to preserve that rank's total base fitness.

The sharing radius σ and objective scaling affect the result.

Exercise 1: MOGA

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Minimize both objectives. B and C have identical objective values.

Candidate	A	B	C	D	E	F	G	H
f_1	1	2	2	5	8	4	6	7
f_2	8	6	6	3	1	7	6	8

1. List the dominators of each candidate and calculate r(x).

2. Assign position scores 8, 7, …, 1, then average within each rank.

3. Given niche counts n(A) = n(D) = n(E) = 1 and n(B) = n(C) = 2, calculate shared and rescaled fitness for rank 1.

Solution: dominance ranks

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Candidate	Dominated by	Count	MOGA rank
A, B, C, D, E	None	0	1
F	B, C	2	3
G	B, C, D	3	4
H	A, B, C, D, F, G	6	7

For F: B = (2, 6) and C = (2, 6) both dominate (4, 7).

Ranks need not be consecutive: ranks 2, 5 and 6 are absent.

Solution: base fitness

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Rank	Candidates	Position scores	Base fitness each
1	A, B, C, D, E	8, 7, 6, 5, 4	6
3	F	3	3
4	G	2	2
7	H	1	1

For rank 1: (8 + 7 + 6 + 5 + 4) / 5 = 6.

Equal rank gives equal base fitness before sharing.

Solution: fitness sharing

Rank 1 has total base fitness $5 \times 6 = 30$.

Candidates	Base F_i	Niche count n_i	Shared F'_i	Rescaled fitness
A, D, E	6	1	6	7.5 each
B, C	6	2	3	3.75 each

Shared total $= 3 \times 6 + 2 \times 3 = 24$.

Rescaling factor $= 30 / 24 = 1.25$.

A, D and E each receive twice the selection weight of B or C.

NSGA-II: algorithm

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Non-dominated Sorting Genetic Algorithm II

1. Initialize N parents P_0, then assign front ranks and crowding distances.
2. Select parents, apply crossover and mutation, and evaluate N offspring Q_t.
3. Combine parents and offspring: $R_t = P_t \cup Q_t$, with size 2N.
4. Sort R_t into fronts and calculate crowding distances within each front.
5. Select N survivors to form P_{t+1}. Repeat steps 2–5.

Non-dominated sorting

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1. F_1 contains every individual with no dominator in the current set.

2. Remove F_1. The non-dominated individuals that remain form F_2.

3. Continue until every individual belongs to a front.

NSGA-II rank = front index.

Fast implementation tracks each individual's dominator count and the set of individuals it dominates.

Crowding distance

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Within one front, sort individuals separately by each objective.

$$CD(i) = \Sigma_j \, [f_j(i^+) - f_j(i^-)] \, / \, [\max f_j - \min f_j]$$

i^- and i^+ are the adjacent individuals in the ordering for objective j.

Give boundary individuals infinite distance (∞).

Larger crowding distance favours a less crowded region.

Use the range within this front. Skip an objective with zero range.

Selection in NSGA-II

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Parent selection

Binary tournament:
1. Prefer the lower front rank.
2. For equal ranks, prefer larger CD.
3. Break a remaining tie randomly.

Survivor selection

Accept complete fronts in rank order.
When a front does not fit, take its largest-CD individuals until the population contains N survivors.

Exercise 2: NSGA-II

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Minimize both objectives. Population size N = 5.

Parents P_t = {A, B, C, D, E} Offspring Q_t = {F, G, H, I, J}

Candidate	f_1	f_2
A	1	5
B	5	1
C	2	10
D	3	8
E	4	7

Candidate	f_1	f_2
F	7	4
G	10	2
H	5	9
I	8	8
J	9	6

All candidates are feasible. Use the combined set $R_t = P_t \cup Q_t$.

Exercise 2: selection

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1. Sort the ten candidates into non-dominated fronts.

2. Determine how many places remain after accepting complete fronts.

3. Calculate crowding distance for every candidate in the splitting front.

4. Select the five survivors and justify the last selected candidate.

Use the objective range within each front.
Assign infinite distance to boundary candidates.

Solution: Pareto fronts

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Front	Candidates	Reason
F_1	A, B	Neither has a dominator in R_t.
F_2	C, D, E, F, G	Non-dominated after removing F_1.
F_3	H, I, J	Non-dominated after removing F_1 and F_2.

Accept A and B. Three places remain.

F_2 has five candidates, so crowding distance decides which three survive.

Solution: crowding distance

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Splitting front: C(2,10), D(3,8), E(4,7), F(7,4), G(10,2)

Both objective ranges are $10 - 2 = 8$.

Candidate	f_1 contribution	f_2 contribution	Total CD
C	Boundary	Boundary	∞
D	(4 − 2) / 8	(10 − 7) / 8	5/8 = 0.625
E	(7 − 3) / 8	(8 − 4) / 8	8/8 = 1.000
F	(10 − 4) / 8	(7 − 2) / 8	11/8 = 1.375
G	Boundary	Boundary	∞

The f_2 ordering is G, F, E, D, C. Each contribution uses its own neighbours.

Solution: survivors

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Selection stage	Accepted	Places left
Complete front F_1	A, B	3
Boundary candidates in F_2	C, G	1
Largest remaining crowding distance	F (CD = 1.375)	0

$$P_{t+1} = \{A, B, C, F, G\}$$

E and D have lower crowding distances. No candidate in F_3 survives.

MOGA and NSGA-II

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Mechanism	Classic MOGA	NSGA-II
Quality ranking	1 + number of dominators	Non-dominated front index
Diversity	Fitness sharing	Crowding distance
Density parameter	Sharing radius σ	No sharing radius
Survivors	Basic generational replacement	Best N from parents + offspring

For Exercise 1, F and G have MOGA ranks 3 and 4, but both lie in the second non-dominated front.

Key conclusions

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1. A MOP can have several useful trade-off solutions.
2. Pareto dominance compares solutions without fixed objective weights.
3. MOGA uses dominance counts and fitness sharing.
4. NSGA-II uses front ranks, crowding distance and elitist survival.

A non-dominated population is an approximation, not proof of global optimality.

References

Fonseca, C. M., and Fleming, P. J. (1993).

Genetic Algorithms for Multiobjective Optimization:
Formulation, Discussion and Generalization.
Proceedings of the Fifth International Conference on Genetic Algorithms,
416–423.

Deb, K., Pratap, A., Agarwal, S., and Meyarivan, T. (2002).

A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II.
IEEE Transactions on Evolutionary Computation, 6(2), 182–197.
DOI: 10.1109/4235.996017

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Thanks for your time

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